[Total No. of Pages - 01]

[Total No. of Question -04]

G H Raisoni International Skill Tech University, Pune

T.Y .B.Tech (TERM -I)

CAE I- Winter Term 2026 (2025 Pattern)

25BTAIPEL504C: Optimization Techniques in Machine Learning

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	B
Q1.(a)	What is optimization? Explain its role in machine learning.	[5]	CO1	
(b)	What is feature scaling? Explain its importance in machine learning.	[5]	CO1	
	OR			
Q 2.(a)	Explain various evaluation metrics used for feature selection.	[5]	CO1	
(b)	What is the need for feature selection in machine learning?	[5]	CO1	
Q 3. (a)	What is filter-based feature selection? Explain with an example.	[5]	CO2	
(b)	What are wrapper-based feature selection methods? Explain with examples.	[5]	CO2	
	OR			
Q. 4 (a)	Explain Correlation-based Feature Selection (CFS) with an example.	[5]	CO2	
(b)	Discuss the advantages and disadvantages of wrapper methods.	[5]	CO2	

----------ALL THE BEST----------